

Project RFQ: Unified Stock Scanner Automation System (Revised V2 - January 30, 2026)

Project Overview

This RFQ outlines a modular, automated personal research system for TSX/TSXV (with future USA exchange support) consisting of two core modules:

1. Insider Trading Scanner (mirroring strategies from "Wall Street Vision Trading Mastery" eBook – Chapters 1-2, relevant filters/back testing sections).
2. Technical Scan Module (MGPR scoring for short-term upside under \$2/\$5/\$10 CAD thresholds).

Python handles data fetching, processing, filtering, calculations, ratings, back testing, and data pushes. Airtable provides the shared dashboard. Zapier is incorporated as a modular no-code layer for workflows, alerts, and optional trading signals. Trading automation (TradingView webhooks + Questrade API bridge) is provided as an optional, toggleable Phase 2 extension with strict manual review/approval safeguards.

Key Principles:

- Personal research tool only – not financial advice. Include prominent disclaimers in code, outputs, emails, dashboards ("This is not financial advice; conduct your own due diligence. Past performance ≠ future results.").
- Modularity: Configurable filters (env vars/config file), linked data for cross-referencing insider + technical signals, easy expansion (USA exchanges, additional modules).
- Security: Env vars for all keys (Polygon, Airtable, Zapier webhook, Gmail, Questrade OAuth). No hard-coded credentials. Sandbox/testing mode mandatory for trading.
- Data Flow: Python (nightly pre-market) → Airtable API (primary) + optional Zapier webhook → Airtable dashboard + alerts/trading signals.
- References: Original specs from "Project RFQ Working copy 20260120 V1.docx". Zapier/TradingView/Questrade integrations based on current 2026 capabilities (TradingView webhooks, Questrade REST API with OAuth 2.0).

Deliverables & Requirements (Common to Both Proposals)

- Full documentation (README.md + PDF guide): Setup, env vars, run instructions, field mappings, testing steps.
- Unit/integration tests (key functions, edge cases, sample data).
- Sample datasets & mock runs.
- Mobile-friendly, clean UI.
- Performance: Optimized (batch calls, pandas vectorized ops). Schedule: Cron/AWS Lambda free tier (~7 AM EST/5 AM MST insider, 7:30 AM MST technical).

The project is divided into two independent but integrated proposals. Bidders may quote on one or both; coordination between specialists required for API key sharing and field alignment.

Proposal 1: Airtable Dashboard Specialist

Scope

Focus exclusively on Airtable base creation, structure, views, interfaces, and Zapier-ready setup. No Python coding or data processing.

Core Deliverables

- Base Creation: New base ("InsiderTechDashboard" or similar). Provide Base ID, table names, field IDs/types, and Personal Access Token (with scoped permissions) to client for Python/Zapier sharing.
- Tables (precise field types to match Python JSON payloads):
 - Raw Insider Data
 - Market Pulls (shared OHLCV, ATR%, volume, etc.)
 - Filtered/Qualified Insider List (ticker link, rating 1-10, SL/TP currency, reasons, ATR%, volume, season variant, gap note, simulation results)
 - Technical Scans Under \$5/\$10/\$20 CAD (ticker, full name, description, current price CAD, 52w high, MGPR 0-100 + breakdown, entry/SL/TP, chart signals, rationale, catalysts)
 - Historical/Back test (shared metrics)
- Linked Records & Formulas: Links between tables (e.g., qualifiers ↔ raw + market pulls). Rollups (avg rating, counts), formulas (e.g., ATR% derivations).
- Views: Filtered grids/galleries/kanban for daily top picks (Insider sorted by rating desc.; Technical segmented by threshold, top 5-8 by MGPR). Combined overview with summaries.
- Interfaces (Airtable Interfaces feature):
 - Daily Dashboard (tabs: Insider, Technical \$5/\$10/\$20; expandable records, charts/summaries).
 - Historical Tracking (filters by date/module).
 - Automation Prep View: Records flagged for Zapier (e.g., high rating + "Enable Alert" checkbox).
- Zapier Integration Prep:
 - Fields: "Zapier Alert Enabled" (checkbox), "Auto-Trade Eligible" (checkbox, default off), "TradingView Webhook Payload" (long text or formula-generated JSON).
 - Views optimized as Zapier triggers (e.g., "New Qualified Stocks" filtered view).
 - Native Airtable automations (e.g., auto-archive >30 days) + notes for Zapier triggers ("New Record in Qualified tables → Send email/Slack or POST to webhook").
- Disclaimers & Best Practices: Dedicated Notes table/view/banner in interfaces. Encryption, access controls, rate-limit awareness. Import/test with sample CSVs.
- Documentation: Base structure guide, API integration notes for Python dev (endpoints, field IDs), Zapier setup examples.

Skills Required: Airtable expert (tables, interfaces, formulas, links), Zapier familiarity for trigger prep.

Proposal 2: Python & Automation Developer (Balance of Work)

Scope

All Python scripting, data logic, Zapier integration, alerts, back testing, and optional trading automation. Airtable push is primary; Zapier is modular extension.

Core Deliverables

- Data Sources & Processing (modular classes: InsiderProcessor, TechnicalScanner, MarketDataHandler):
 - Insider: Scrape OpenInsider.com (requests/BeautifulSoup, retries/logging) or fallback SEC EDGAR. Key fields: ticker, dates, insider details, shares/price/value.
 - Technical: Polygon API (OHLCV, indicators via ta-lib or manual numpy/pandas: ATR, RSI, MACD, beta, etc.) + web scraping/news for catalysts. Scan TSX/TSXV (future: up to 3 USA exchanges configurable).
 - Shared Market Data: Polygon aggregates/bars, ATR% calculation, volume checks, SPY gap logic.
- Filters & Logic (configurable via env vars/config):
 - Insider: ATR% variants, volume ranges, season-based SL/TP, multiples/repeat checks, SPY gap note.
 - Technical: Beta, RSI/MACD crossover (ta-lib), volume surge, MGPR scoring (weighted sub-scores: Trend Strength, Momentum, Volatility, Volume → 0-100 total + qualitative ratings; normalize with linear/sigmoid). Generate rationales, chart signals (candlestick patterns via ta-lib), entry/SL/TP.
 - Shared: pandas DataFrames, error handling, logging. Simulate ratings/reasons (e.g., "Multiples + earnings = 9/10").
- Automation & Outputs:
 - Main script (main.py) + modules. Daily run (cron/Lambda). Temporary JSON/CSV → push to Airtable tables via API (requests POST, match field types).
 - Zapier Integration: Optional toggle (env var USE_ZAPIER=true). POST processed payloads (ticker, rating, SL/TP, JSON for TradingView) to Zapier webhook URL. Zapier handles enrichment, retries, routing (e.g., Airtable update fallback, email/Slack).
 - Alerts: smtplib Gmail (HTML email: sections for insider + technical segments, Matplotlib simple charts as attachments if feasible) or Slack webhook. Configurable recipient. Include disclaimers.
- Back testing Module (separate backtest.py/function): Historical simulation ("buy open/sell close" or short-term for technical), win rate, returns, Sharpe, drawdown, vs. random control. Use Polygon historical or sample CSVs. Target edge per eBook.
- Trading Automation Extension (toggleable, default OFF; Phase 2):
 - Generate TradingView-compatible JSON payload (action: buy/sell, ticker, entry, SL/TP) and POST via Zapier webhook or direct from Python.

- Questrade Bridge: Use Questrade REST API (OAuth 2.0, sandbox first) for order placement (bracket orders with SL/TP). Include approval flow (e.g., Airtable "Approved for Auto-Trade" checkbox → Zapier/Python checks before execute). Position sizing/risk limits configurable. Portfolio tracking push back to Airtable.
 - Start semi-automated (alerts + manual review); evolve to full auto only after testing/small positions. Use TradersPost-like bridges if needed for no-code (custom webhook intermediary recommended if no native Questrade Zapier support).
- Rating Systems: Insider 0-100 (similar to MGPR: Implement as a function with weighted sub-scores – e.g., Insider Strength (25%: multiples, buy type), Volatility (25%: ATR match), Liquidity (25%: volume sweet spot, no spikes), Timing (25%: earnings season, no repeat, gap context) → normalize each 0-25 via linear scaling or sigmoid, sum to 0-100 total + qualitative ratings (e.g., 80-100 Excellent, 60-79 Good, 40-59 Fair,
- Security/Testing/Docs: Env vars, exceptions, unit tests (MGPR, filters, ratings), full run simulations, README + user guide PDF.
- Skills Required: Python (requests, bs4, pandas, numpy, ta-lib, matplotlib), API integrations (Polygon, Airtable, Zapier webhook, optional Questrade OAuth), scheduling (cron/Lambda), secure coding.